



9/1/2026

N + - Network Fundamentals & Building Networks.

Firewalls NAT and Packet Filtering.

TASK 1.

❖ Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.

➤ Windows Defender Firewall Inbound Rule:

Procedure:

- Opened **Windows Defender Firewall with Advanced Security**.
- Created a new **Inbound Rule** for **TCP port 8080**.
- Selected **Block the connection** and enabled the rule.
- Named the rule **“Block Incoming Port 8080”**.

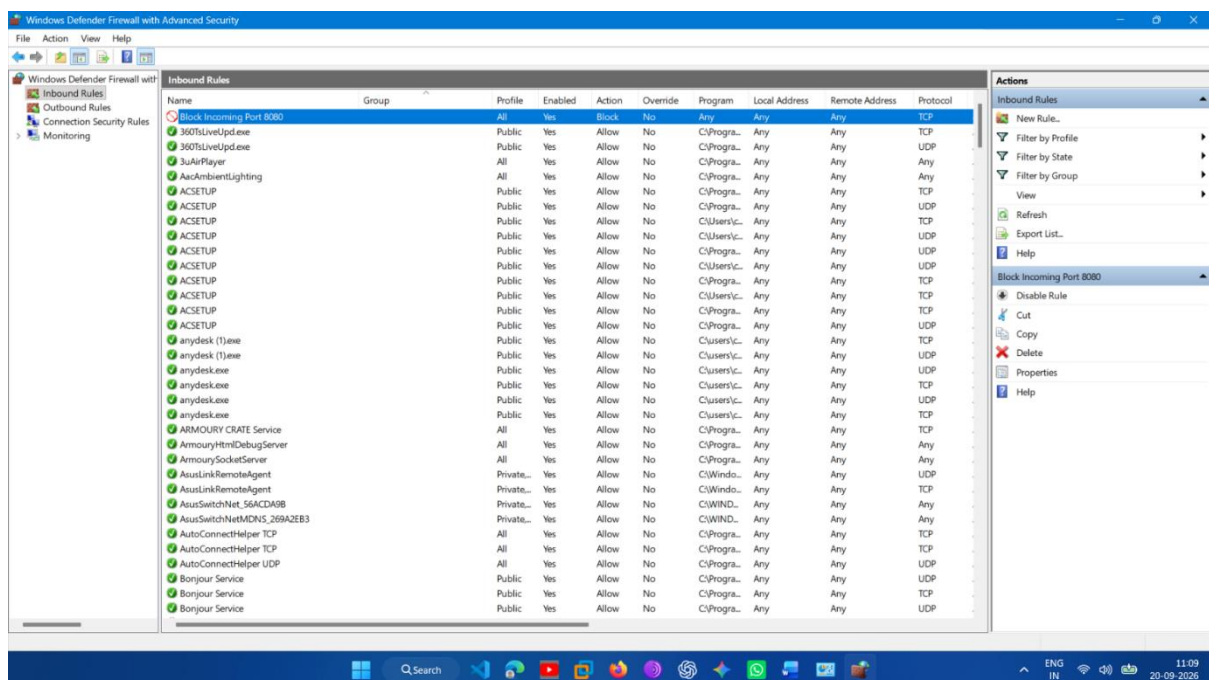


Figure : Block Incoming Port 8080 Rule.

Result: Incoming traffic on **port 8080** is blocked successfully.

TASK 2.

❖ Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access, Use any drawing tool or even hand-draw and click a photo.

➤ NAT (Network Address Translation):

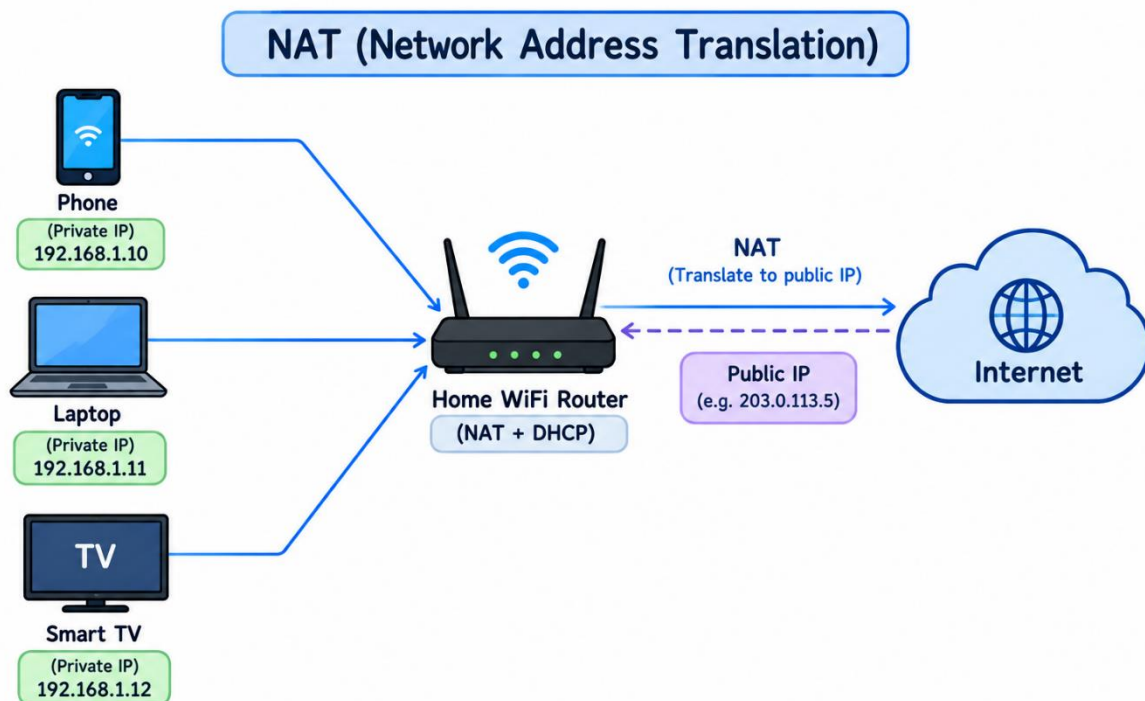


Figure : NAT Network Diagram.

Explanation:

- ✓ The **Wi-Fi router assigns private IP addresses** to the phone, laptop, and smart TV inside the home network. When these devices access the Internet, the router uses **NAT to translate their private IP addresses into one common public IP address**. This allows all three devices to access the Internet through the same public IP.

TASK 3.

- ❖ **Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.**

➤ Configuration:

Steps:

- ➦ Open the **Wi-Fi router's Admin Panel**.
- ➦ Go to **Port Filtering / Firewall Settings**.
- ➦ Create a new rule to **block outgoing traffic on TCP port 25 (SMTP)**.
- ➦ Save and apply the settings.

Explanation:

- ✓ Blocking port 25 prevents devices on the network from directly sending emails through the standard SMTP port. This can help reduce **spam emails sent by malware or compromised devices** on the network.

TASK 4.

❖ Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53, Focus on source/destination IPs and ports in the rule set.

➤ Packet Filtering:

Packet	Source → Destination	Port	Result
1. HTTP	192.168.1.5 → 172.217.166.78	80	Allowed
2. SSH	10.0.0.10 → 34.201.23.1	22	Blocked
3. DNS	192.168.1.7 → 8.8.8.8	53	Allowed

Note: These results assume the sample rule set allows **HTTP** and **DNS** but blocks **SSH**.

TASK 5.

❖ Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

➤ Use of Firewalls and Packet Filtering:

Block unwanted traffic: Instagram or Zomato can use firewalls to block unauthorized IP addresses, ports, and suspicious network requests from reaching their servers.

Control access: Packet filtering can allow only required types of traffic, such as HTTPS on port 443, while blocking unnecessary or risky connections.

Prevent misuse and attacks: Firewalls can detect and restrict unusual traffic patterns or repeated connection attempts, helping protect servers from attacks and misuse.